

SoMaSLAM: 2D Graph SLAM for Sparse Range Sensing and Soft Manhattan World Constraints



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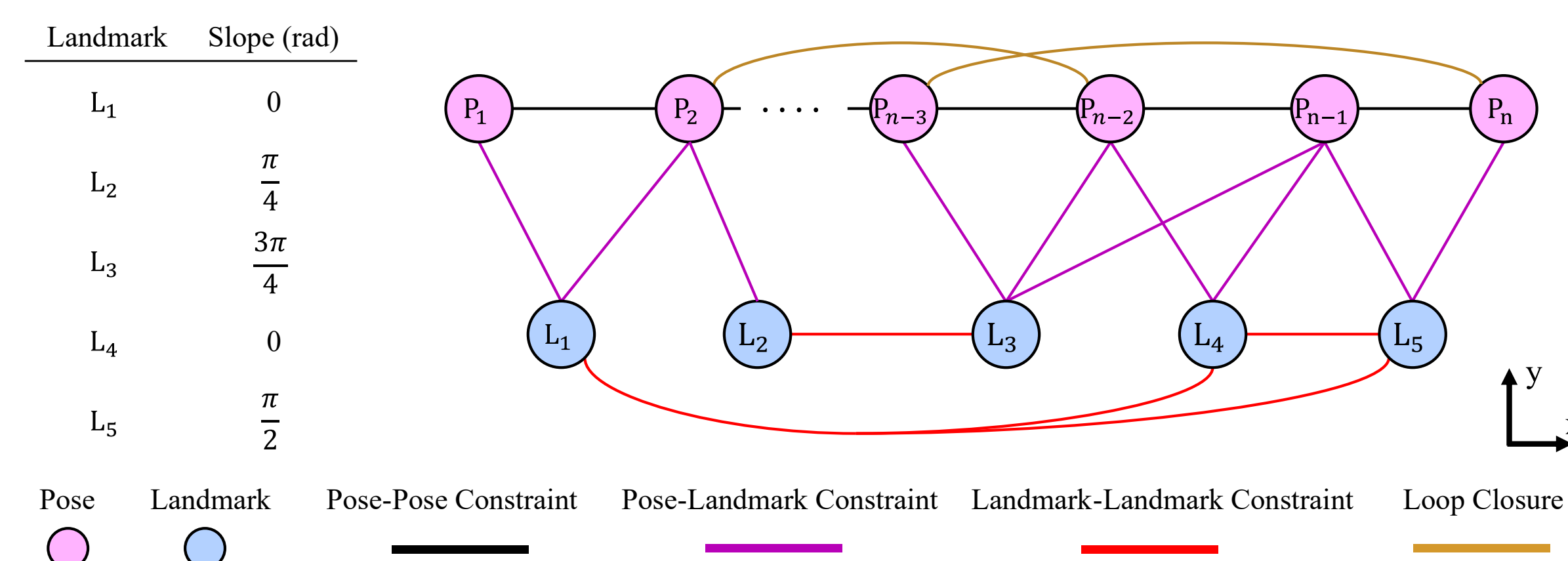


Introduction & Motivation

- Nano drones (Crazyflies) have a 15g payload capacity and limited sparse sensors
- Minimal data points accumulate trajectory drift
- Strict priors (Manhattan World assumptions) fail when walls are not perfectly Manhattan



Core Idea: Soft Manhattan World



- Encourage, rather than force, nearby walls to suit Manhattan World conditions (soft Manhattan World)
- Introducing landmark-landmark constraints to the factor graph framework

Key Formulation

Landmarks should be local, geometrically significant, and non-saturated

Variable error

$e_l(l_1, l_2) = \theta_{2,ideal} - \theta_2$

$\theta_{2,ideal}$: ideal orientation of landmark 2

“ideal”: it suits the Manhattan World assumption

Length-weighted information matrix

$$\Omega = \begin{bmatrix} \text{len}_i + \text{len}_j & 0 \\ 0 & \text{len}_i + \text{len}_j \end{bmatrix}$$

Adjacency matrix between landmark i and landmark j

Longer, stable walls get higher priority over shorter walls

Constraint formulation

$$G(X) = \sum_{ij} e_l(l_i, l_j)^T \Omega_i e_l(l_i, l_j) + F(X)$$

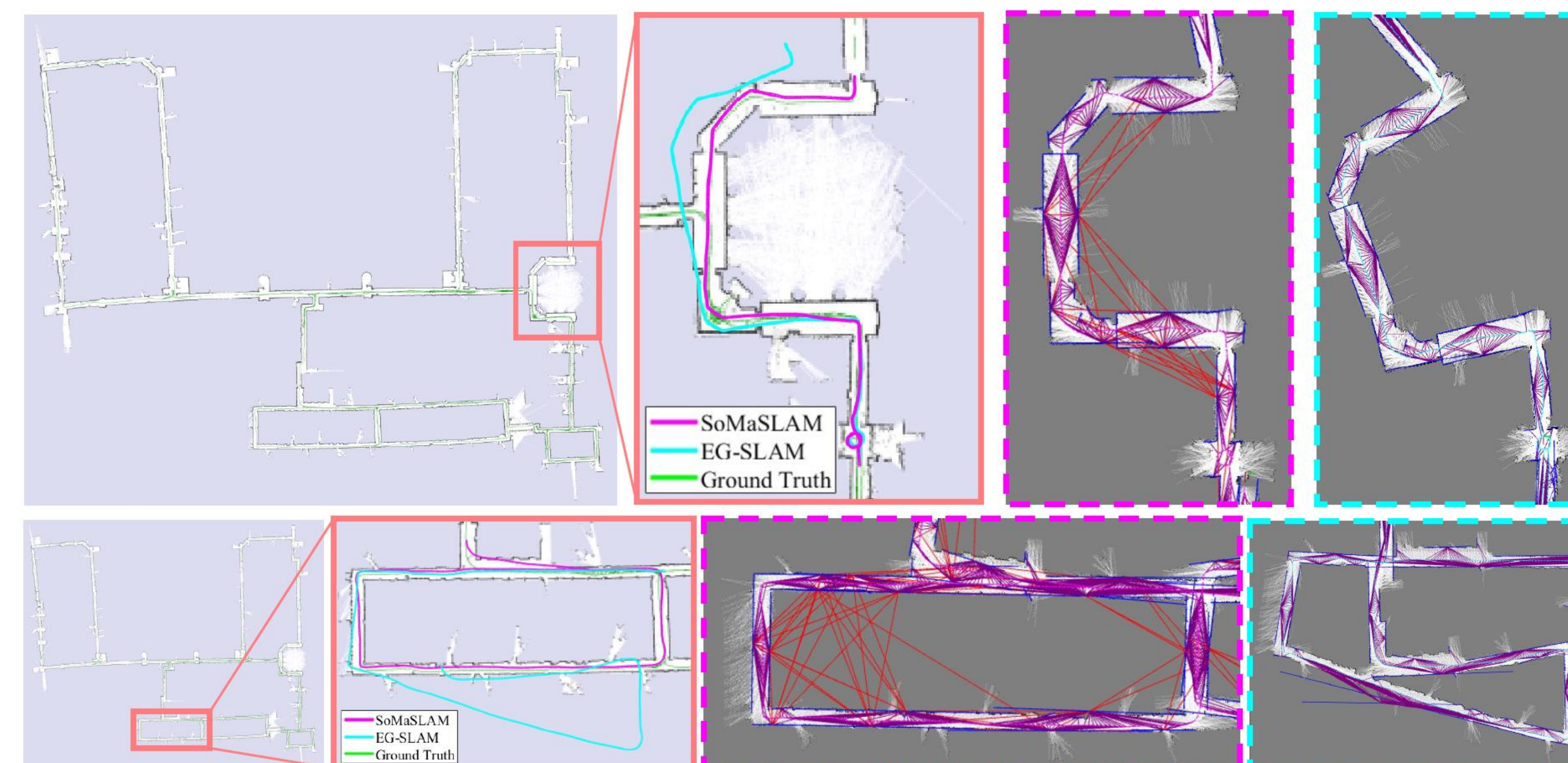
Soft Manhattan World constraint logic

Classic SLAM optimization by EG-SLAM

$$F(X) = \sum_i \|e_o(\mathbf{x}_i, \mathbf{x}_{i+1})\|_{\Sigma_i}^2 + \sum_{ij} \|e_l(\mathbf{x}_i, \mathbf{l}_j)\|_{\Sigma_{ij}}^2$$

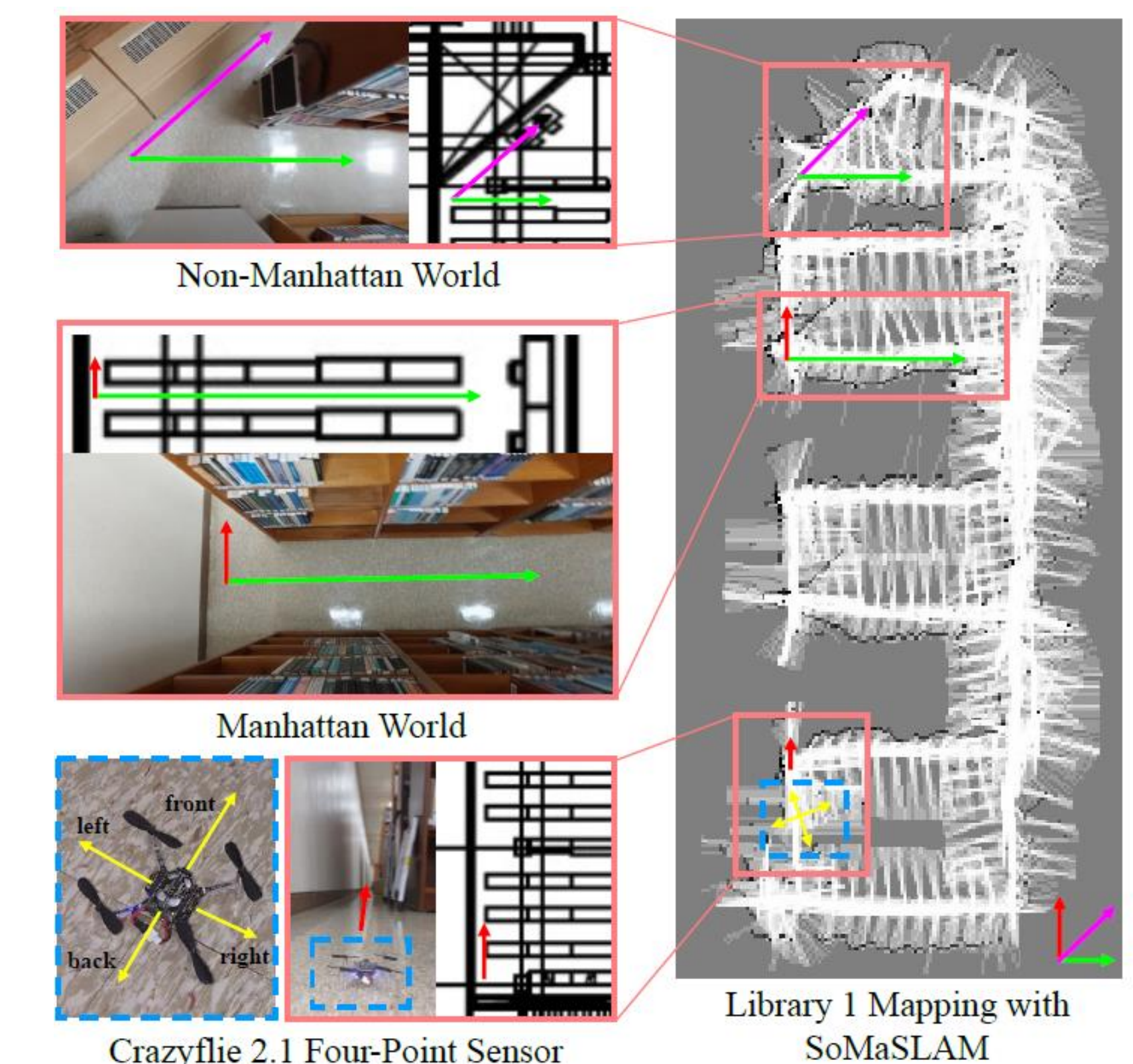
Experimental Results

Dense public datasets



	SoMaSLAM	EG-SLAM	FL-SLAM
Aces			
Abs. Trans. (m)	0.04 ± 0.05	0.04 ± 0.05	0.55 ± 2.68
Abs. Rot. (°)	1.04 ± 1.34	1.04 ± 1.36	2.33 ± 6.51
Intel Lab			
Abs. Trans. (m)	0.13 ± 0.21	2.80 ± 8.98	7.37 ± 14.35
Abs. Rot. (°)	2.71 ± 3.01	15.19 ± 36.00	39.17 ± 53.80
MIT Killian			
Abs. Trans. (m)	0.91 ± 4.49	17.84 ± 46.25	50.32 ± 128.15
Abs. Rot. (°)	2.19 ± 3.77	15.81 ± 37.69	9.26 ± 27.03
MIT CSAIL			
Abs. Trans. (m)	0.39 ± 1.24	2.53 ± 9.00	1.68 ± 5.29
Abs. Rot. (°)	4.10 ± 5.68	11.45 ± 28.23	5.07 ± 8.24
Freiburg-079			
Abs. Trans. (m)	0.28 ± 0.72	0.52 ± 1.18	6.50 ± 13.67
Abs. Rot. (°)	3.86 ± 6.23	4.77 ± 6.16	30.29 ± 45.53

Sparse custom Crazyfly datasets



Datasets	SoMaSLAM	EG-SLAM	FL-SLAM	Ground Truth
Library 1				
Library 2				

The gist: landmark-landmark constraints in factor graphs!